POLICY FRAMEWORK · FEBRUARY 2026

A Global Framework for Sustainable AI

A synthesis of the world's most effective environmental regulations and standards, applied to artificial intelligence across its full lifecycle — from raw material extraction to hardware end-of-life.

SCOPE
Global · All Inhabited Continents

LIFECYCLE COVERAGE
7 Phases · Full Chain

REGULATORY SOURCES
60+ Instruments

FRAMEWORK PILLARS
Measure · Disclose · Reduce

415 TWh Global data center electricity 2024	**662%** Real emissions vs. officially reported	**80–90%** AI compute used for inference
945 TWh IEA data center projection by 2030	**180 Mt** Annual CO_2 from data centers	**1,125M m³** Projected US AI water footprint by 2030

Preface

Artificial intelligence is already a global technology, and its environmental footprint is already a global problem. Data centers consumed an estimated 415 terawatt-hours (TWh) of electricity in

2024, roughly the entire national consumption of Indonesia, and the International Energy Agency (IEA) projects that figure could more than double to 945 TWh by 2030. AI-specific servers are responsible for a growing share of that load, with their water consumption, carbon emissions, hardware waste, and raw material demand compounding the picture further.

Crucially, existing sustainability regulation has not caught up. The world has spent decades building sophisticated environmental frameworks, from the Paris Agreement to ISO 14001 to the EU's Corporate Sustainability Reporting Directive, but virtually none were designed with AI in mind. Meanwhile, the AI Act, the most comprehensive attempt at AI governance to date, substantially diluted its own environmental provisions during final negotiations, leaving the industry largely self-regulated on sustainability.

This framework argues that we do not need to start from scratch. The world already possesses a rich inventory of proven environmental laws, reporting systems, lifecycle standards, and governance mechanisms. What is missing is their synthesis and deliberate application to AI's unique characteristics: its global supply chains, its exponential scaling, its opacity, and its dual role as both a major environmental consumer and a potentially powerful tool for climate solutions.

The goal of this framework is to fill that gap by drawing from the best of what exists globally and constructing a coherent, actionable system for making AI development and deployment genuinely sustainable across its entire lifecycle.

Part 1: Foundations

1.1 The Scale of AI's Environmental Footprint

Energy and Carbon. Global data centers currently account for approximately 180 million tonnes (Mt) of indirect CO_2 emissions annually from electricity consumption. Google's carbon emissions rose approximately 48% over five years largely due to AI and cloud computing; Microsoft's rose 23.4% since 2020. Training a single large language model can consume around 50 gigawatt-hours (GWh) of electricity. Critically, inference — the ongoing use of deployed models — may ultimately generate more total emissions than training, as popular models are queried millions or billions of times per day.

Water. Research published in Nature Sustainability projects that AI server deployment in the US alone could generate an annual water footprint of 731 to 1,125 million cubic meters between 2024 and 2030. Globally, by 2025, AI systems could consume as much water as all bottled water consumed worldwide in a year. Yet major tech companies currently do not publish AI-specific water use figures, and the EU AI Act does not address water consumption at all.

Hardware and Raw Materials. Each GPU required for AI training or inference carries a substantial embodied carbon cost from manufacturing, material extraction, and transport. An estimated 3.85 million GPUs were shipped to data centers in 2023, up from 2.67 million in 2022. The mining of rare earth elements and other materials used in AI hardware involves toxic processes and significant land disruption. GPU lifespans are shortening as models demand more powerful hardware, accelerating the generation of electronic waste.

The Opacity Problem. Major AI companies do not publish AI-specific emissions or water use data. Aggregate "carbon neutral" claims obscure actual local emissions. Data center emissions are estimated to be up to 662% higher than what major tech companies report, primarily due to methodological choices around market-based versus location-based accounting.

1.2 AI's Dual Role: Consumer and Potential Solution

This framework does not treat AI as inherently harmful or inherently beneficial. It treats AI as a technology whose sustainability outcomes depend entirely on how it is built, governed, and used. AI applications in energy optimization, climate modeling, materials science, agricultural efficiency, and grid management have documented potential to reduce emissions far beyond AI's own footprint — if deployed intentionally and responsibly. The framework's goal is to maximize the latter while systematically reducing the former.

1.3 Why Existing Frameworks Fall Short

The GHG Protocol, ISO 14001, the Paris Agreement, and the EU's CSRD are proven, widely adopted systems. Their limitation is not their quality; it is their scope. They were not designed for technologies that scale exponentially faster than reporting cycles, supply chains that span semiconductor fabs in Taiwan, training clusters in Virginia, and inference servers in Singapore simultaneously, or industries where the largest actors have no legal obligation to disclose the relevant data.

1.4 The Full AI Lifecycle

1. **Raw Material Extraction:** mining of rare earth elements, lithium, cobalt, and other materials for hardware
2. **Hardware Manufacturing:** GPU, TPU, and chip fabrication; semiconductor production; server assembly
3. **Data Collection and Storage:** acquisition, processing, and storage of training data
4. **Model Training:** compute-intensive development of AI models
5. **Model Deployment and Inference:** ongoing operation, real-world use, and query processing

6. **Maintenance and Updates:** ongoing fine-tuning, retraining, and system maintenance
7. **Hardware End-of-Life:** decommissioning, recycling, and disposal of physical infrastructure

Part 2: Global Regulatory Inventory

This section synthesizes the most effective and relevant environmental laws, standards, reporting systems, and governance mechanisms from around the world, organized by the dimension of sustainability they address. The full inventory covers 60+ instruments across all inhabited continents across six dimensions: emissions and climate, energy efficiency, water, hardware lifecycle and electronic waste, biodiversity and land use, and transparency and governance.

2.1 Selected Key Instruments by Region

EU European Union

- EU AI Act (2024/1689) — energy documentation for GPAI
- Corporate Sustainability Reporting Directive (CSRD)
- Energy Efficiency Directive 2023/1791
- Corporate Sustainability Due Diligence Directive (CSDDD)
- EU Taxonomy Regulation
- Ecodesign for Sustainable Products (ESPR)
- WEEE Directive / RoHS Directive

🌐 International

- Paris Agreement / UNFCCC — NDCs, net-zero targets
- GHG Protocol (WRI/WBCSD) — Scope 1, 2, 3
- IFRS S1 / S2 (ISSB Standards)
- ISO 14001, 50001, 42001
- ISSA 5000 Assurance Standard
- Basel Convention / Kimberley Process
- Alliance for Water Stewardship Standard

🌎 Latin America

- Brazil CVM Resolutions 217–219 (IFRS S1/S2)
- Brazil Sustainable Taxonomy (TSB)
- Brazil AI Plan (PBIA 2024–2028)
- Mexico Sustainable Taxonomy (2023)
- Chile Extended Producer Responsibility (REP) Law
- Escazú Agreement (24 countries)
- Colombia Environmental Crimes Law

🌏 Asia-Pacific

- Singapore Green Data Centre Roadmap (2024)
- China GEC Policy / East Data West Computing
- South Korea Framework Act on AI
- Japan FSA Sustainability Standard (ISSB-aligned)
- Australia ISSB-aligned Disclosures
- ASEAN Digital Economy Framework

🌍 Africa

- African Union Agenda 2063
- AU Climate Change Strategy 2022–2032
- South Africa Climate Change Act (2024)
- South Africa Carbon Tax Phase 2 (2026)
- Kenya Climate Change Act
- Nigeria Climate Change Act
- Morocco NDC (CCPI #9 globally)

US United States

- California SB 253 — Scope 1, 2, 3 disclosure
- California SB 261 — Climate-related risk disclosure
- US AI Environmental Impacts Act of 2024 (proposed)
- EPA study on AI environmental impacts
- NIST AI environmental metrics development

2.2 Critical Regulatory Gap: Water

> **MOST URGENT GAP**
>
> No major jurisdiction currently mandates AI-specific water use disclosure. The EU AI Act does not address water at all. The EU EED requires data center water reporting but does not distinguish AI workloads and does not apply outside EU member states. This framework identifies water as the most urgent regulatory gap requiring new policy.

Part 3: The Framework

3.1 Framework Architecture

The framework is organized around three interlocking elements: three pillars, seven lifecycle phases, and three compliance tiers.

📏 **Measure**

Standardized metrics for all environmental dimensions across the AI lifecycle. Carries 25% of each phase score.

📣 **Disclose**

Mandatory, verified, and comparable reporting requirements. Carries 30% of each phase score.

📉 **Reduce**

Binding improvement targets tied to best available technology. Carries 45% of each phase score.

3.2 The Cascading Model

No single jurisdiction has produced a comprehensive AI sustainability standard, but the components of one already exist — distributed globally. The cascading model synthesizes these into a unified standard by identifying the most ambitious, evidence-based requirement for each issue area and applying it globally.

> **LAYER 1 — BASELINE**
>
> Draws from binding laws already in force: EU EED annual reporting, Brazil IFRS S1/S2 disclosures from 2026, California Scope 1/2/3 disclosure (SB 253), EU AI Act energy documentation for GPAI models.

> **LAYER 2 — CONVERGENCE STANDARD**

Draws from the most advanced jurisdictions: Singapore PUE of 1.3 or lower, China 80% renewable for national hub data centers by 2030, South Africa legally binding sectoral emissions targets, Kenya geothermal data center model.

LAYER 3 — FRAMEWORK PROPOSALS

Addresses gaps where no regulation yet exists: Water Protocol equivalent to GHG Protocol; Cumulative Lifecycle Emissions Accounting (CLEA) for inference; Hardware Circularity Passport; AI Minerals Certification Scheme; Leapfrog Incentive Pathway; AI Climate Benefit Verification Standard.

3.3 Differentiated Responsibilities Model

Inspired by the UNFCCC's Common But Differentiated Responsibilities principle, the framework defines three organizational categories. All actors share a common responsibility to minimize AI's environmental footprint. The pace, depth, and specific mechanisms of compliance differ based on organizational scale, geographic context, and historical contribution to the problem.

Category A
Frontier Developers & Hyperscale Operators

- →24/7 carbon-free energy matching
- →Full lifecycle water disclosure
- →Third-party verified emissions (ISSA 5000)
- →Supply chain due diligence to point of extraction
- →Hardware Circularity Passports
- →Immediate compliance with all stringent requirements

Category B
Enterprise AI Deployers

- →Tier 1 compliance immediately
- →Pathway to Tier 2 within 3 years
- →Pathway to Tier 3 within 5 years
- →Scope 3 AI emissions in sustainability reports
- →Sustainability criteria in vendor contracts
- →Model right-sizing assessments

Category C
SMEs, Researchers & Emerging Economies

- →Tier 1 compliance in the near term
- →Technical assistance and capacity building
- →Green financing mechanisms
- →Leapfrog Incentive Pathway available
- →Priority for hardware reuse programs
- →Support from international institutions

3.4 The Master Matrix

The matrix below maps all seven lifecycle phases plus governance against three pillars, with tier and category designations. Each cell summarizes the core requirement at the most demanding tier for which a regulatory precedent exists (Layer 1 or 2) or which the framework proposes (Layer 3).

Phase	📏 Measure	📣 Disclose	📉 Reduce
⛏ Materials (8%)	Scope 3 upstream emissions; water contamination risk; conflict mineral flag	Supply chain due diligence reports; TNFD nature disclosures (T1, Cat A/B)	Zero sourcing from non-compliant suppliers; AI minerals certification scheme (T3)
🔧 Hardware Mfg (10%)	Embodied carbon per GPU/TPU; fab water use; PFAS tracking	Scope 1/2/3 disclosure; RoHS/WEEE compliance declarations (T1, all)	Min. lifespan standards; modularity; near-zero embodied carbon targets (T3, Cat A)
💾 Data Storage (7%)	kWh/TB; PUE/WUE; data redundancy ratio (DER)	Storage energy as line item; data minimization policy (T1, Cat A/B)	Data minimization; cold storage migration; deduplication mandates (T2, all)
🧠 Training (22%)	Energy per run (MWh); grid carbon intensity; WUE; FLOPs/kWh	Training Run Environmental Card; ISSA 5000 verified (T2–T3, Cat A)	≥80% renewable (T1); 24/7 CFE matching (T3); advanced/underground cooling
⚡ Inference (25%)	Joules per query; carbon per query; CLEA rolling cumulative records	Quarterly per-query intensity; annual CLEA public reporting (T2–T3)	Carbon-aware routing; quantization/distillation; year-on-year efficiency targets
🔄 Maintenance (8%)	Energy per retraining run; retraining vs. fine-tuning delta	Update energy as lifecycle report line item (T1, Cat A/B)	Efficient Update Preference policy; model deprecation schedule (T2, all)
♻️ End-of-Life (10%)	Hardware lifespan; e-waste (tonnes); recycling recovery rate	WEEE compliance; e-waste destination audit; Circularity Passport (T2, Cat A/B)	Take-back programs; Basel-compliant disposal; refurbishment pathways (T1–T3)
📋 Governance (10%)	GHG Protocol-aligned AI metrics; measurement frequency	Third-party verified (ISSA 5000); IFRS S2/CSRD aligned (T2–T3, Cat A)	CSO with AI mandate; board oversight; binding reduction targets (T2, Cat A)

3.5–3.11 Lifecycle Phase Summaries

Phase 1: Raw Material Extraction

Weight: 8%

The framework proposes an AI Hardware Minerals Certification scheme for cobalt, coltan, lithium, and rare earths — modeled on the Kimberley Process. The Congo Basin region receives highest-priority due diligence requirements given its irreplaceable carbon sink value.

Phase 2: Hardware Manufacturing

Weight: 10%

The framework extends EU ESPR logic to AI hardware, requiring minimum lifespan standards, right-to-repair provisions, modular design, mandatory chemical transparency declarations, and PFAS phase-out timelines for semiconductor fabrication.

Phase 3: Data Collection & Storage

Weight: 7%

The framework introduces data minimization as a sustainability principle, drawing from GDPR logic and applying an environmental dimension. Two new metrics: the Data Efficiency Ratio (DER) and Storage Carbon Intensity (SCI).

Phase 4: Model Training

Weight: 22%

A mandatory Training Run Environmental Card is proposed for every publicly deployed model. 24/7 carbon-free energy (CFE) matching is the Tier 3 standard; annual RECs are Tier 1 minimum. Underground siting and immersion cooling are Tier 3 best practice.

Phase 5: Deployment & Inference

Weight: 25%

Inference represents 80–90% of AI's total compute yet is almost entirely absent from existing regulation. Cumulative Lifecycle Emissions Accounting (CLEA) is the framework's most significant original regulatory proposal. Carbon-intensity-aware workload routing is a Tier 2 requirement for Category A actors.

Phase 6: Maintenance & Updates

Weight: 8%

Efficient Update Preference establishes that fine-tuning is the sustainability-preferred approach over full retraining where performance parity can be achieved. Model Deprecation Policy requires lifecycle policies with sustainability as an explicit criterion.

Phase 7: Hardware End-of-Life

Weight: 10%

The Hardware Circularity Passport is a digital record accompanying every piece of AI hardware from manufacture through decommissioning. Refurbishment and cascade reuse is the preferred end-of-life pathway wherever hardware has remaining useful life.

Governance & Transparency

Weight: 10%

Cross-cutting dimension covering third-party verified reporting, GHG Protocol alignment, board-level sustainability oversight, and the institutional mechanisms that make all other pillars enforceable rather than aspirational.

3.12 Three Compliance Tiers

Tier 1

Minimum Compliance

Score range: 41–60/100

- All baseline legal requirements met
- Annual renewable energy certificates
- Basic energy and emissions reporting
- WEEE take-back compliance
- GHG Protocol Scope 1 and 2 disclosure

Tier 2

Good Practice

Score range: 61–75/100

- Hourly CFE matching in same grid region
- WUE targets with closed-loop cooling
- Training Run Environmental Cards published
- Inference energy in sustainability reports
- Supply chain due diligence to Tier 2

Tier 3

Best Practice

Score range: 76–100/100

- 24/7 carbon-free energy matching (verified)
- CLEA publicly reported quarterly
- Third-party assurance under ISSA 5000
- Hardware Circularity Passports in use
- Full supply chain traceability to extraction

3.13 Original Framework Proposals (Layer 3)

Training Run Environmental Card

A standardized disclosure document accompanying every publicly deployed model. Includes total energy consumed, grid carbon intensity at training time, water consumed, hardware used and its embodied carbon, and a cumulative lifecycle emissions projection based on anticipated deployment scale.

Regulatory anchor: EU AI Act Article 40 and GHG Protocol product lifecycle accounting

Cumulative Lifecycle Emissions Accounting (CLEA)

Requires AI companies to maintain and publicly report rolling cumulative emissions accounts per deployed model — tracking training emissions plus all inference emissions as they accumulate over the model's operational life. Addresses the single largest gap in existing AI sustainability governance.

Regulatory anchor: GHG Protocol product lifecycle guidance, extended to inference phase

Hardware Circularity Passport

A digital record accompanying every piece of AI hardware from manufacture through decommissioning, tracking its environmental profile, operational history, and end-of-life destination. Enables accurate Scope 3 accounting, verified recycling, and Basel Convention enforcement.

Regulatory anchor: EU Digital Product Passport concept under ESPR

AI Minerals Certification Scheme

A dedicated supply chain certification scheme for cobalt, coltan, lithium, and rare earth elements used in AI hardware — modeled on the Kimberley Process but designed for AI-specific mineral supply chains, particularly the Congo Basin region.

Regulatory anchor: Kimberley Process certification logic, OECD Due Diligence Guidance

Leapfrog Incentive Pathway

Organizations in countries without reliable renewable grid access that commit to and achieve renewable-native AI infrastructure receive accelerated compliance timelines and higher-tier framework certification, directly incentivizing the bypass of fossil-fuel data center models in emerging economies.

Regulatory anchor: CBDR principles from UNFCCC, adapted for private sector

AI Climate Benefit Verification Standard

A methodology for quantifying, attributing, and verifying emissions reductions enabled by AI applications in climate-relevant sectors — grid optimization, climate modeling, precision agriculture, materials science. Creates positive incentives for deploying AI in service of environmental outcomes.

Regulatory anchor: Verra and Gold Standard voluntary carbon market methodologies

Part 4: Implementation Guidelines

Part 4 provides tiered, staged guidance for six actor groups across three horizons: immediate actions (Year 1), a transition pathway (Years 2–5), and maturity (Year 5+).

AI Developers and Model Providers (Category A)

Year 1: Full environmental audit of model portfolio; implement Training Run Environmental Card; begin supply chain mapping to flag high-risk DRC-region sourcing.
Years 2–5: Embed compute efficiency targets into model development approval; transition from annual RECs to 24/7 CFE matching; publish model deprecation policies; set water reduction targets.
Maturity: CLEA operational and publicly reported quarterly; all training at 24/7 CFE; full supply chain due diligence; Hardware Circularity Passports; ISSA 5000 verified reporting annually.

Enterprise AI Deployers (Category B)

Year 1: Inventory all AI usage by provider; begin Scope 3 inference emissions estimates; establish sustainability criteria in AI vendor procurement.
Years 2–5: Model right-sizing assessments; transition over-specified deployments to more efficient alternatives; apply EU GPP sustainability criteria to AI vendor contracts.
Maturity: Full Scope 3 visibility over AI-related emissions; sustainability standards in all vendor contracts; AI environmental performance reported as a material business risk.

Data Center Operators

Year 1: Ensure EED/Singapore/China compliance; install disaggregated AI versus non-AI energy metering; conduct water source assessment.
Years 2–5: Roadmap from annual RECs to 24/7 CFE; specify advanced cooling for new builds; identify waste heat off-takers.
Maturity: PUE of 1.2 or lower; WUE approaching zero evaporative loss; 100% 24/7 verified CFE; ISO 50001 certified; new facilities built to subsurface or equivalent cooling specification.

Hardware Manufacturers

Year 1: Publish embodied carbon disclosures for primary AI hardware lines using standardized LCA; publish annual fabrication water use data disaggregated by facility.
Years 2–5: Redesign product lines for longevity and modularity; implement global take-back programs; pilot Hardware Circularity Passport for flagship lines.
Maturity: Full catalog embodied carbon published; PFAS phased out; Hardware Circularity Passports on all products; certified mineral supply chains.

Governments and Regulators

Legislate AI-specific inference energy disclosure — the single largest regulatory gap. Extend EED-equivalent reporting to all major AI markets with extraterritorial application. Develop interoperable AI sustainability standards through NIST, BSI, DIN, and equivalent bodies. Add AI sustainability as a permanent UNFCCC COP agenda item. Establish Green AI Infrastructure Fund providing concessional financing for renewable-powered data center development in low- and middle-income countries.

Investors and Financial Institutions

Year 1: Integrate AI-specific environmental metrics into ESG assessment frameworks; engage data providers to develop AI-specific scoring consistent with this framework.
Years 2–5: Shareholder engagement pushing AI companies toward framework compliance; green loan frameworks conditioning favorable terms on framework progress.
Maturity: Full integration of AI sustainability into investment analysis; portfolio-level AI environmental impact data published; AI criteria embedded in green bond and transition finance frameworks.

Part 5: Gaps and Future Recommendations

Eight critical gaps are identified, ordered by urgency, defined by the scale of ungoverned environmental impact and the feasibility of closing the gap with near-term action.

1 **Water** Critical

No major jurisdiction mandates AI-specific water use disclosure. Three actions needed: develop a Water Protocol parallel to the GHG Protocol; amend the EU EED to require AI-disaggregated indirect water data; elevate WUE to mandatory reporting globally with regional water stress multipliers applied in high-stress zones.

2 **Inference Emissions** Critical

Inference represents 80–90% of AI's compute but is almost entirely absent from regulation. Adopt CLEA as mandatory reporting for systems receiving more than 1 million queries per month; extend EU AI Act Article 40 to cover inference; require cloud AI providers to supply per-query inference energy data to customers.

3 **The Open-Source Exemption** High

EU AI Act exemptions for open-source models mean widely deployed models like Meta's Llama family face no sustainability reporting requirements. Restructure (not eliminate) the exemption: disclosure-only obligations at release time for providers above a compute threshold; distribute ongoing inference responsibility to deployers.

4 **Hardware Supply Chain Accountability** High

DRC-region cobalt and coltan sourcing involves documented environmental degradation. PFAS contamination from semiconductor fabrication is largely unaddressed. Actions: AI Minerals Certification Scheme via OECD; RoHS PFAS phase-out timelines; Basel Convention enforcement for AI hardware; mandatory Hardware Circularity Passport under ESPR by 2028.

5 **Greenwashing and Claim Integrity** High

Data center emissions are estimated at 662% higher than officially reported due to annual REC methodologies. Require mandatory third-party verification under ISSA 5000 for Category A actors within 3 years; mandate disclosure of both market-based and location-based emissions figures where they differ materially.

6 **AI's Positive Environmental Potential** Medium

No framework creates incentives for deploying AI in service of verified environmental benefits. Develop an AI Climate Benefit Verification Standard through a multi-stakeholder process involving AI developers, climate scientists, and voluntary carbon market bodies.

7 **Regulatory Capacity in Emerging Economies** High

Africa's data center market grows at 11.79% annually; Malaysia's demand could reach 30% of national electricity by 2030; India's capacity projected to grow from 1.4 GW to 9 GW by 2030. Establish international capacity-building programs, the Green AI Infrastructure Fund, and support regional regulatory cooperation bodies.

8 **Absence of International Governance Architecture** Critical

No international body has a mandate for AI sustainability standards; no treaty creates binding AI environmental obligations. Three actions ordered by feasibility: add AI sustainability to UNFCCC COP as a permanent workstream; develop OECD binding guidelines on AI sustainability; work toward a dedicated multilateral instrument on AI and the environment.

CONCLUSION: THE WINDOW FOR ACTION

The regulatory vacuum around AI sustainability is not permanent — it is a window. The technology is new enough that its governance architecture is still being designed, the standards are still being written, and the institutional structures are still being built. The decisions made in the next five years will determine whether AI becomes one of the forces that accelerates the climate crisis or one of the forces that helps address it. The right direction is clear: measure AI's full environmental footprint across its complete lifecycle, disclose it transparently and with verified accuracy, reduce it against science-based targets with differentiated timelines that account for capacity and equity, and build the international governance architecture that makes all of this binding rather than voluntary. The tools largely exist. The regulatory precedents largely exist. What has been missing, until now, is their synthesis into a coherent, global, AI-specific framework.

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